

# Automated Recognition Model for Identifying Harmful and Harmless Insects in Crop Management

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**Abstract**—Diseases and pests often affect agriculture, causing significant economic losses. Worldwide, pests can cause yield losses of up to 40%. To minimize these losses, detecting and identifying pests as early as possible is crucial. Prior studies have developed detection models to detect harmful or harmless insects. However, there has been no model developed to detect both categories together. To address this issue, we aim to develop a model that can detect and classify both harmful and harmless insects in agricultural environments. We will assess the accuracy of three different methods: YOLO (You Only Look Once) versions 8 and 9, and VGG16 (Visual Geometry Group) on a dataset comprising ten classes, five for harmful insects and five for harmless insects, to determine the most effective approach. The results indicate that YOLOv9 achieved the highest accuracy of 0.972, followed closely by YOLOv8 with 0.969, while VGG16 lagged at 0.849. This suggests that YOLOv9 is the most effective tool among the tested models for detecting and classifying both harmful and harmless insects in agricultural settings.

**Keywords**—YOLO, VGG-16, Insects, Harmful, Harmless

## I. INTRODUCTION

In agricultural settings, insects play crucial roles, both beneficial and harmful. Beneficial insects, such as bees and butterflies, aid in pollination, whereas pests like gypsy moths, codling moths, and diamondback moths cause significant crop damage by feeding on foliage, leading to substantial yield losses [1]. Traditionally, identifying these insects for effective farm management relied on manual methods, which were time-consuming and error-prone due to the vast number of insect species.

Advancements in Artificial Intelligence (AI), particularly in machine learning, have revolutionized pest detection. This paper compares three advanced AI models—YOLOv8, YOLOv9, and VGG16—which utilize convolutional neural networks

(CNNs) for automated insect classification. YOLO models are renowned for their speed and accuracy in real-time object detection [2], while VGG16 is noted for its deep-layer architecture that excels in classifying and recognizing complex image features [3].

The goal of this study is to develop a model capable of accurately detecting and classifying both harmful and harmless insects in agricultural environments using advanced AI techniques. This objective meets the critical need for precise identification to support targeted pest management, minimize crop damage, and promote sustainable farming practices.

This research assesses these models' effectiveness in recognizing various insect species to support the agricultural industry's need for sustainable farming technologies that enhance crop yields. It includes a thorough review of existing research, details on data collection and model training, and an evaluation of each model's performance under real-world conditions. The paper concludes with recommendations for further research and potential improvements in AI-driven pest management solutions.

## II. LITERATURE REVIEW

This section reviews the literature related to detection models for identifying harmful and harmless insects. The research is organized chronologically, from 2023 to 2018.

Diniati Ruaika improved the accuracy of the ResNet50 model for butterfly classification using data augmentation and ensemble deep learning, achieving 95% accuracy on a Kaggle dataset of 75 butterfly species [4].

Kumar developed a YOLOv5-based model for insect detection, achieving a 93% mean average precision and an F1 score

of 0.90. They proposed future enhancements by integrating MobileNet with YOLOv5x for faster real-time applications [5].

Sorbelli applied RGB cameras and drones in orchards to detect the ornate brown stink bug, using machine learning models that demonstrated satisfactory performance with the YOLO framework [6].

Stark utilized YOLO models to recognize pollinators, training on over 17,000 images and achieving up to 97% accuracy. They highlighted the potential for further improvements in complex environments [7].

Anwar and Masood developed an ensemble model using pre-trained networks like VGG16, achieving 82.5% accuracy and suggesting enhancements with YOLO and Faster RCNN for better performance in diverse classifications [8].

Wen introduced Pest-YOLO, which uses drones and modified YOLO frameworks to detect agricultural pests, achieving a mean average precision of 69.59% and recall of 77.71% [9].

Thenmozhi Kasinathan automated crop pest detection using machine learning, with CNN models achieving up to 91.5% accuracy [10]. They discussed integrating these algorithms into deeper CNN models for larger datasets.

Marković developed a model using environmental sensors to predict pest appearance with 86.3% accuracy, emphasizing the importance of early detection for effective pest management [11].

Homchan used the Google Teachable Machine tool to classify cricket species, achieving 100% accuracy with a model trained on over 2,000 images [12].

AI utilized Inception-ResNet-v2 to identify crop diseases across 10 crops with an 86.1% accuracy, while also developing a WeChat applet for real-time pest and disease identification [13].

Deng employed saliency maps and various feature extraction methods to identify insect pests, achieving an 85.5% accuracy rate and highlighting the potential for practical agricultural applications [14].

The advancement of automated pest detection and management in agriculture has been significantly enhanced by machine learning and deep learning techniques. However, **notable gaps remain** that need to be addressed for improved efficacy. While several studies have utilized YOLO models for pest detection, few have compared the latest versions (YOLOv8 and YOLOv9) or addressed the classification of both harmful and harmless insects. This distinction is crucial for targeted pest management and minimizing crop damage. Therefore, this study aims to fill these gaps by developing a model using YOLOv8, YOLOv9, and VGG16 to effectively classify pests based on their impact on crop health. This approach seeks to achieve higher accuracy, faster processing, and the critical differentiation between pest types.

### III. DATA COLLECTION

We collected a dataset from Kaggle, specifically designed for agricultural environments, featuring ten categories of frequently encountered insects. This includes a split between

harmful insects, such as Marmorated Stink Bugs, Colorado Potato Beetles, Fall Armyworms, Western Corn Rootworms, and Thrips; and harmless insects like Bees, Beetles, Cicadas, Dragonflies, and Grasshoppers. Each category in the dataset is represented by 100 images, totaling 1,000 images. These images are organized into training, validation, and testing sets, with 80%, 10%, and 10% of the images allocated, respectively, resulting in 800 training images and 100 each for validation and testing. The complete dataset is available for access in the designated drive folder [15].

## IV. METHODOLOGY

### A. Training Methodology

The purpose of this study is to compare previous research findings with new discoveries related to the identification of beneficial and harmful insects on farms using the You Only Look Once (YOLO) and VGG-16 object recognition algorithms such as Table I [16], [17]. The study evaluates the effectiveness of the Yolo8l, Yolo9-e, and VGG-16 models, which were trained on a dataset of 1000 images distributed across 10 classes, with each class containing 100 images [18], [19]. The models were trained using different hyperparameters, including epochs ranging from 40 to 100 and batch sizes of 8 for Yolo8 and Yolo9, and 10 for VGG-16.

The YOLOv8 model was trained across various epochs: 40, 50, 70, and 100, with a batch size of 8 and a default learning rate of 0.01. Similarly, the YOLOv9 model underwent training for the same epoch configurations (40, 50, 70, and 100) with the same batch size and learning rate. The VGG-16 model, on the other hand, was trained for 35 and 40 epochs, employing a batch size of 10 and a learning rate of 0.01.

TABLE I  
HYPERPARAMETERS USED FOR TRAINING YOLOv8, YOLOv9, AND VGG-16 MODELS

| Model    | Hyperparameters | Value  |
|----------|-----------------|--------|
| YOLOv8   | Epochs          | 40-100 |
|          | Batch size      | 8      |
| YOLOv9-e | Epochs          | 40-100 |
|          | Batch size      | 8      |
| VGG-16   | Epochs          | 35-40  |
|          | Batch size      | 10     |

### B. Training Environment

To effectively train our models, Google Colab was utilized for running the Python code. This platform provides advanced computational capabilities, including access to GPUs, which are essential for meeting the specific training demands of both the YOLO and VGG-16 models. The environment specifications included the use of NVIDIA Tesla T4 GPUs, which significantly accelerated the training process.

## V. RESULTS AND DISCUSSION

In this section, we analyze and compare the performance of three advanced models (YOLOv8, YOLOv9, and VGG-16) based on accuracy, precision, recall, and F1-score across different insect classes.

### A. YOLO Object Detection and Classification

#### • YOLOv8 performance

Based on Table II, we tested the YOLOv8 model over various epochs and observed that it achieved accuracies of 81%, 80%, 83%, and 81% for epochs 40, 50, 70, and 100, respectively. The highest accuracy was found at epoch 70, which was 83%. We will now proceed to calculate the precision, recall, and F1-score for this epoch to further evaluate the model's performance.

TABLE II  
YOLOv8 ACCURACY OVER EPOCHS

| Model  | Epochs | Accuracy |
|--------|--------|----------|
| YOLOv8 | 40     | 0.81     |
|        | 50     | 0.80     |
|        | 70     | 0.83     |
|        | 100    | 0.81     |

We use standardized formulas to measure the performance of our YOLO detection models. These metrics are calculated based on values from the confusion matrix:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1 Score} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (4)$$

These calculations help us effectively evaluate the performance of our predictive models in distinguishing between different classes.

Table III details the YOLOv8 model's performance at epoch 70, highlighting its accuracy, precision, recall, and F1-scores across various insect classes. The model records an overall accuracy of 96.89%, with Beetles and Brown Marmorated Stink Bugs achieving high F1-scores of 0.927 and 0.904, respectively. Cicadas and Dragonflies also show strong performance with F1-scores of 0.832 and 0.856. However, challenges persist in detecting Fall Armyworms, evident from a low recall of 0.47 and an F1-score of 0.577. Moderate results are noted for Grasshoppers and Thrips, with F1-scores of 0.774 and 0.879. The analysis extends to Colorado Potato Beetles and Western Corn Rootworms, with F1-scores of 0.843 and 0.766, respectively, illustrating a comprehensive evaluation of the model's capabilities at this stage.

TABLE III  
ACCURACY, PRECISION, RECALL, AND F1-SCORE FOR EACH CLASS FOR YOLOv8 AT EPOCH 70

|                             | Accuracy | Precision | Recall | F1-score |
|-----------------------------|----------|-----------|--------|----------|
| Bees                        | 0.975    | 0.884     | 0.84   | 0.862    |
| Beetles                     | 0.986    | 0.905     | 0.95   | 0.927    |
| Cicada                      | 0.969    | 0.816     | 0.848  | 0.832    |
| Dragonfly                   | 0.973    | 0.832     | 0.881  | 0.856    |
| Grasshopper                 | 0.956    | 0.732     | 0.82   | 0.774    |
| Brown Marmorated Stink Bugs | 0.983    | 0.977     | 0.842  | 0.904    |
| Fall Armyworms              | 0.937    | 0.746     | 0.47   | 0.577    |
| Colorado Potato Beetles     | 0.973    | 0.918     | 0.78   | 0.843    |
| Thrips                      | 0.977    | 0.850     | 0.91   | 0.879    |
| Western Corn Rootworms      | 0.960    | 0.818     | 0.72   | 0.766    |

Fig. 1 illustrates the training and validation loss curves and performance metrics for the YOLOv8 model. The x-axes on all plots represent the number of epochs, while the y-axes represent various metrics. 'train/box\_loss,' 'train/cls\_loss,' and 'train/dfl\_loss' represent the training losses for bounding box regression, classification, and Distribution Focal Loss, respectively. Corresponding validation losses are shown as 'val/box\_loss,' 'val/cls\_loss,' and 'val/dfl\_loss.' The metrics 'precision(B)' and 'recall(B)' depict the precision and recall scores, respectively, over epochs, and 'mAP50(B)' and 'mAP50-95(B)' display the mean Average Precision at IoU=0.5 and across different IoU thresholds (0.5 to 0.95). These curves show a consistent decline in losses, indicating effective learning and optimization. The increasing trends in precision, recall, and mAP metrics across different IoU thresholds reflect enhanced detection capabilities and precision in classification. Based on Fig. 2 this visualization summarizes the model's learning progress, demonstrating its improving ability to generalize to unseen data effectively.

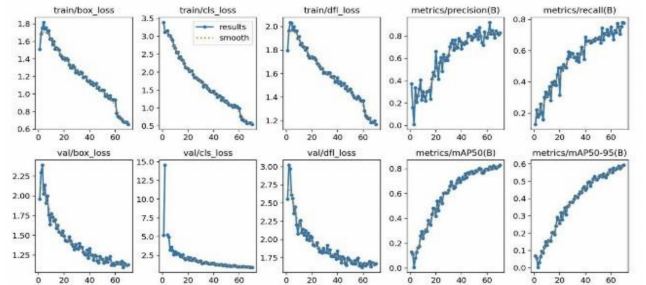


Fig. 1. Training and Validation Losses of YOLOv8 Model

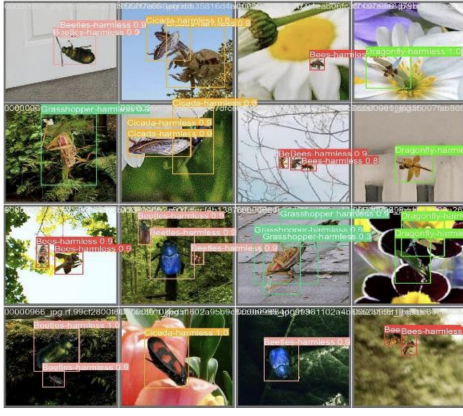


Fig. 2. Testing sample of YOLOv8 Model

Moving from YOLOv8 to YOLOv9, we observe an improved performance profile.

- YOLOv9 performance

In Table IV, we have documented the performance of the YOLOv9 model across different epochs. The model demonstrated accuracies of 85%, 86%, 83%, and 88% at epochs 40, 50, 70, and 100, respectively. Notably, the epoch 100 showed the highest accuracy, reaching 88%. Following this observation, we will next assess the precision, recall, and F1-score corresponding to this optimal epoch.

TABLE IV  
YOLOv9 MODELS ACCURACY AT DIFFERENT EPOCH INTERVALS

| Model  | Epochs | Accuracy |
|--------|--------|----------|
| YOLOv9 | 40     | 0.85     |
|        | 50     | 0.86     |
|        | 70     | 0.83     |
|        | 100    | 0.88     |

TABLE V  
YOLOv9 INSECT DETECTION PERFORMANCE: ACCURACY, PRECISION, RECALL, AND F1-SCORE FOR EACH INSECT CLASS AT EPOCH 100

|                             | Accuracy | Precision | Recall | F1-score |
|-----------------------------|----------|-----------|--------|----------|
| Bees                        | 0.991    | 0.95      | 0.95   | 0.95     |
| Beetles                     | 0.972    | 0.817     | 0.89   | 0.852    |
| Cicada                      | 0.991    | 0.95      | 0.95   | 0.95     |
| Dragonfly                   | 0.980    | 0.899     | 0.881  | 0.890    |
| Grasshopper                 | 0.966    | 0.779     | 0.88   | 0.826    |
| Brown Marmorated Stink Bugs | 0.985    | 1.0       | 0.842  | 0.914    |
| Fall Armyworms              | 0.959    | 0.867     | 0.65   | 0.743    |
| Colorado Potato Beetles     | 0.972    | 0.856     | 0.83   | 0.843    |
| Thrips                      | 0.968    | 0.822     | 0.83   | 0.826    |
| Western Corn Rootworms      | 0.940    | 0.764     | 0.524  | 0.621    |

The YOLOv9 model has shown strong capability in automatically recognizing various insects, achieving high precision, recall, and F1 scores, particularly for Bees and

Cicadas, all at 0.95, as detailed in Table 4. It has excelled with Brown Marmorated Stink Bugs, reaching a perfect precision of 1.0 and an F1-score of 0.914. However, it encounters challenges with Western Corn Rootworms and Fall Armyworms, where F1-scores drop to 0.621 and 0.743, respectively, indicating inconsistent detection. Thrips show moderate performance with an F1-score of 0.826, suggesting areas for improvement. Overall, the model achieves an accuracy of 97.24%, which we obtained after taking the average for the accuracy of each class, reflecting strong performance across insect classes, though further refinements are necessary for optimizing detection in challenging categories.

In Figure 3, you can see the training and validation loss curves of the YOLOv9 model, as well as performance metrics that show how the model is improving over time. The x-axes represent the number of epochs, and the y-axes show various metrics such as 'train/box\_loss,' 'train/cls\_loss,' and 'train/dfl\_loss' for training losses, and 'val/box\_loss,' 'val/cls\_loss,' and 'val/dfl\_loss' for validation losses. The performance metrics 'precision(B),' 'recall(B),' 'mAP50(B),' and 'mAP50-95(B)' are also included. While the validation loss curves may have some fluctuations, they overall indicate good generalization. The performance metrics demonstrate improved accuracy and robustness in object detection tasks, showing that the model effectively learns and generalizes from the training data.

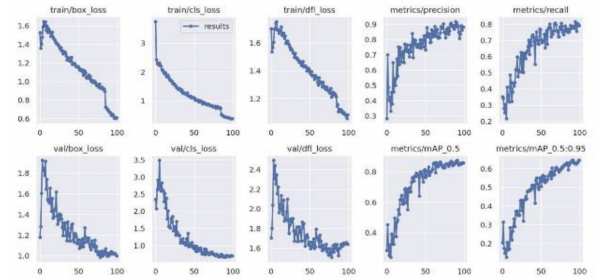


Fig. 3. Training and Validation Losses of YOLOv9 Model

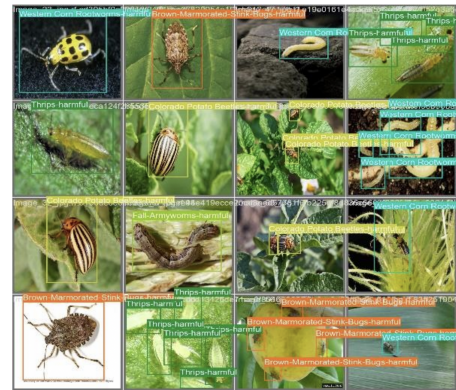


Fig. 4. Testing sample of YOLOv9 Model

- VGG-16 Neural Network Model

After discussing the impressive performance of the YOLOv9 model, we compare it with the VGG-16 model. While it is a well-established model for classical image processing tasks, it may not be as effective in this application. We utilized the previously mentioned formulas to assess the performance of the VGG-16 classification model in Table 6 [20].

Table VI reveals the VGG-16 model's varied performance across insect classes. Bees achieved 87.5% accuracy but only a 24.2% F1-score due to low precision and recall. Beetles and Colorado Potato Beetles recorded similar accuracies around 83% with F1-scores of 29.2%. Cicadas performed the lowest with an 80.7% accuracy and a 17% F1-score. Dragonflies showed an accuracy of 88.5% but a lower F1-score of 20.7%. Grasshoppers and Brown Marmorated Stink Bugs had F1-scores of 10.5% and 12.5%, respectively. Meanwhile, Western Corn Rootworms excelled with a 55% recall and a 40.7% F1-score.

TABLE VI  
ACCURACY, PRECISION, RECALL, AND F1-SCORE FOR EACH CLASS FOR VGG-16

|                             | Accuracy | Precision | Recall | F1-score |
|-----------------------------|----------|-----------|--------|----------|
| Bees                        | 0.875    | 0.308     | 0.2    | 0.242    |
| Beetles                     | 0.839    | 0.25      | 0.35   | 0.292    |
| Cicada                      | 0.807    | 0.148     | 0.2    | 0.170    |
| Dragonfly                   | 0.885    | 0.333     | 0.15   | 0.207    |
| Grasshopper                 | 0.820    | 0.111     | 0.1    | 0.105    |
| Brown Marmorated Stink Bugs | 0.860    | 0.167     | 0.1    | 0.125    |
| Fall Armyworms              | 0.863    | 0.294     | 0.25   | 0.270    |
| Colorado Potato Beetles     | 0.830    | 0.25      | 0.35   | 0.292    |
| Thrips                      | 0.871    | 0.286     | 0.2    | 0.235    |
| Western Corn Rootworms      | 0.840    | 0.324     | 0.55   | 0.407    |

Fig. 5 displays the model's training progression, noting significant improvements after the 20th epoch, particularly in feature recognition and generalization, highlighted by the performance gains in Western Corn Rootworms during later training phases.

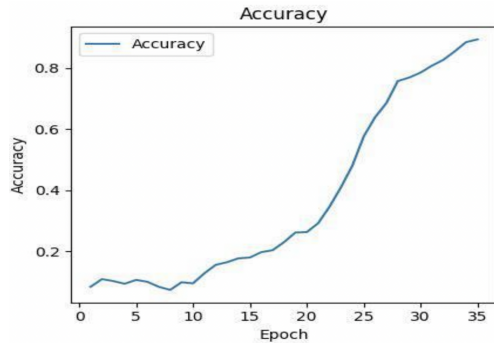


Fig. 5. Training Accuracy of VGG-16 Model

### B. Overall Comparison

The YOLOv9 model significantly outperforms YOLOv8 and VGG-16 in detecting various insects, with an overall accuracy of 97.2%, superior precision, recall, and F1 scores across multiple classes, as detailed in Table VII. It achieves perfect precision for Brown Marmorated Stink Bugs and nearly perfect scores for Bees and Cicadas. Despite its effectiveness, deploying YOLOv9 in real-world agricultural settings poses challenges due to variable environmental conditions, image scale and orientation, motion blur, and the need for continual model updates to adapt to evolving species and conditions.

TABLE VII  
ACCURACY COMPARISON OF YOLOv8, YOLOv9, AND VGG-16  
ACCURACY

| Model  | Accuracy |
|--------|----------|
| YOLOv8 | 0.969    |
| YOLOv9 | 0.972    |
| VGG-16 | 0.849    |

### C. Applicability to Expanding the Dataset

We expanded our dataset to include two new classes: tomato hornworms (harmful) and butterflies (harmless), adding 100 images for each class and increasing the total dataset to 1,200 images across 12 categories. We retrained the YOLOv9 model using this augmented dataset over 100 epochs. This resulted in a slight reduction in mAP50 from 0.88 to 0.87, reflecting the challenges posed by the increased dataset complexity. Nonetheless, the model continued to perform robustly, demonstrating its adaptability and scalability. The successful integration of new classes underscores the model's potential for dynamic agricultural applications, maintaining relevance and effectiveness as new insect classes are encountered.

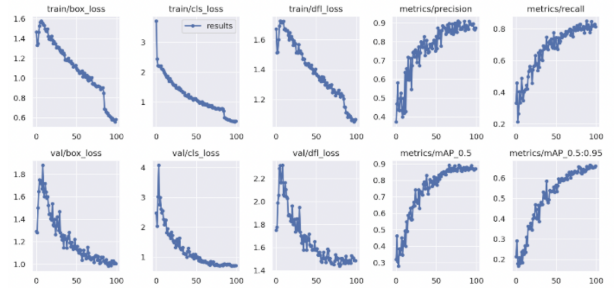


Fig. 6. Training and Validation Losses of YOLOv9 Model after expanding the dataset

### D. YOLOv9 Performance Under Environmental Variations

This section evaluated the YOLOv9 model's resilience under varied environmental conditions, simulating real-world scenarios such as rain, snow, fog, dust storms, nighttime, sun glare, dynamic shadows, reflections, and blurred visibility. We tested the model's performance by introducing these ecological effects into the test images, assessing its ability to detect

objects under different intensities of precipitation, varying densities of fog, low light, glare, and the impacts of dynamic movements.

The YOLOv9 model's resilience was systematically tested against these environmental factors. Results, presented in Fig. 7, demonstrated that the model maintained high detection accuracy across all conditions, proving its effectiveness and robustness in handling real-world environmental variability.



Fig. 7. YOLOv9 model results under various environmental factors

### E. Limitation

In our study, we faced several limitations that could impact the overall effectiveness and applicability of the detection model we developed. Firstly, the dataset we used was limited to specific types of insects. While it covered a variety of harmful and harmless insects, it did not represent the full diversity of insect species present in real-world agricultural settings. This limitation may impact the model's ability to be used in broader applications. Additionally, the dataset did not include images of insects at various life stages, which is crucial for accurate identification and classification in practical scenarios. This absence could make it difficult for the model to recognize insects throughout their lifecycle, limiting its usefulness in comprehensive pest management strategies.

Furthermore, the dataset did not capture images under varied environmental conditions such as rain, snow, fog, dust storms, nighttime, sun glare, dynamic shadows, reflections, and blurred visibility. These factors are often encountered in real-world agricultural environments and can significantly impact the model's performance. We simulated these conditions during testing by photoshopping the images and found that the model performed well, suggesting its adaptability. However, incorporating real-world environmental factors in the training dataset could further enhance the model's accuracy and reliability, ensuring more consistent performance across diverse scenarios.

## VI. CONCLUSION

In conclusion, our study demonstrates the efficacy of advanced object detection models in distinguishing between

harmful and harmless insects in agricultural settings, potentially reducing pest-related economic losses. Comparing YOLO versions 8 and 9 with VGG-16, we found YOLOv9 to be the most accurate, with a 0.972 accuracy rate. This highlights YOLOv9's advantages for real-time, accurate pest detection. While VGG-16 showed lower accuracy, it remains useful for applications not requiring the highest precision. Future work should refine these models, expand the dataset to include more insect classes, and develop a user-friendly tool for farmers. Advancing these technologies can enhance pest management and protect crop yields more effectively.

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